

Genomics review for HCC1395

Files used for pVACview/ITB review:

~/hcc1395-immuno/final_results/pVACseq/mhc_i/HCC1395_TUMOR_DNA.all_epitopes.aggregated.mhci.tsv
~/hcc1395-immuno/final_results/pVACseq/mhc_i/HCC1395_TUMOR_DNA.all_epitopes.aggregated.metrics.mhci.json
~/hcc1395-immuno/final_results/pVACseq/mhc_ii/HCC1395_TUMOR_DNA.all_epitopes.aggregated.mhcii.tsv

Files used for genomics review:

[gs:// path to Reference Bucket or path to local reference files]/Homo_sapiens.GRCh38.105.sorted.coding.gtf
~/hcc1395-immuno/final_results/annotated.expression.vcf.gz
~/hcc1395-immuno/final_results/normal.cram
~/hcc1395-immuno/final_results/tumor.cram
~/hcc1395-immuno/final_results/maseq/alignments/MarkedSorted.bam
~/hcc1395-immuno/manual_review/hcc1395.igv_session.xml

Introduction

This is a hypothetical case used for demonstration purposes only, based on data obtained from the breast cancer cell line HCC1395 and a matched normal lymphoblastoid cell line. This example assumes the target delivery platform is synthetic long peptide vaccine. The sequencing was performed at Washington University. The tumor type is breast. Data analysis was performed using the WASHU immuno.wdl pipeline on the Google Cloud Platform (run time: 2 days, 13 hours, 4minutes).

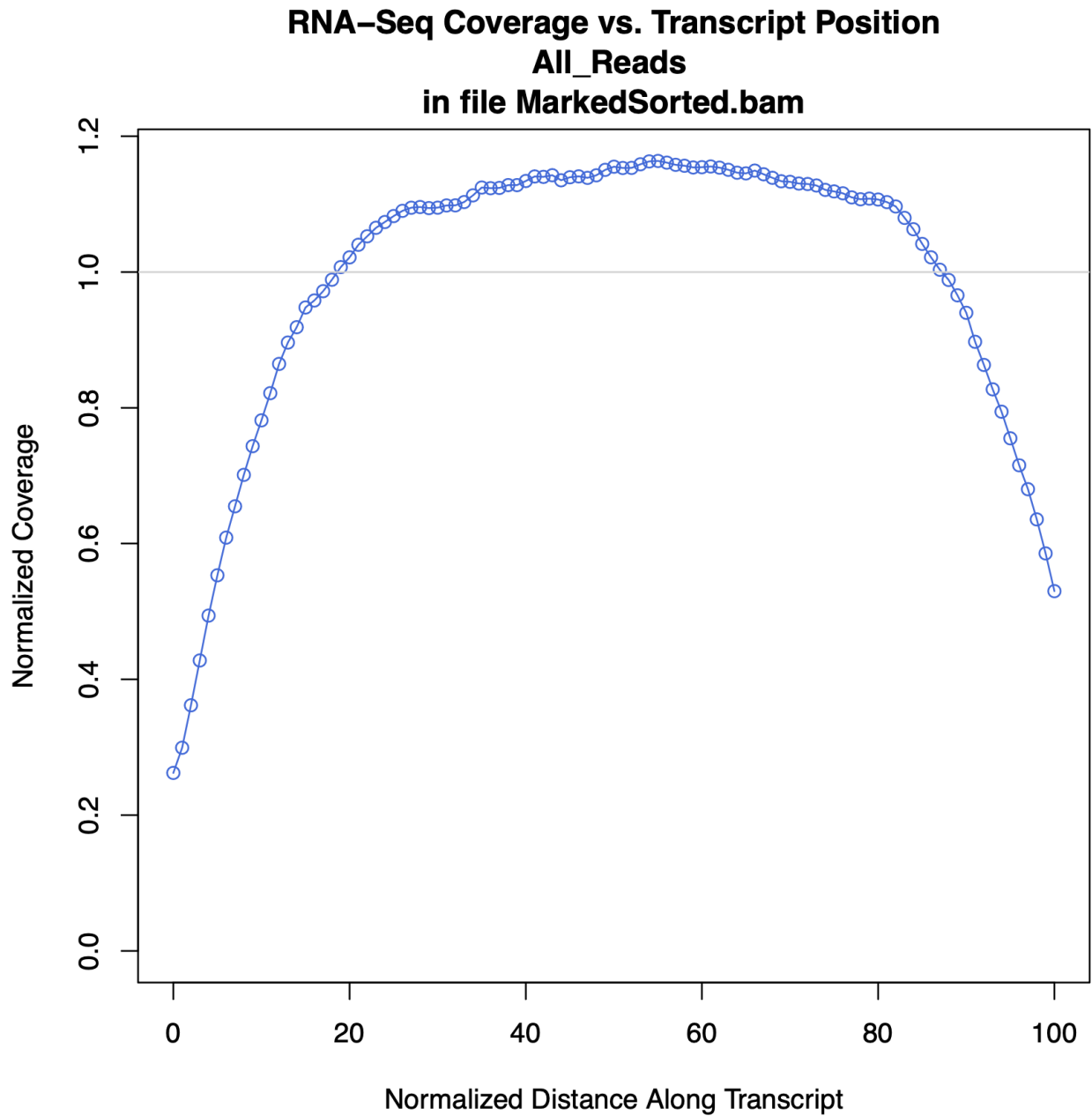
Costs:

totalCost: \$47.03
diskCost: \$2.22
cpuCost: \$32.10
memoryCost: \$12.71
workflowId: f09bfc78-b446-47b4-9c3b-e652596991d0

Basic data QC review

- normal_dna_aligned_metrics.txt Unique Map Reads: **137,753,212 (excellent)**
- tumor_dna_aligned_metrics.txt Unique Map Reads: **170,023,028 (excellent)**
- tumor_rna_aligned_metrics.txt Unique Map Reads: **129,905,103 (excellent)**
- normal_dna_aligned_metrics.txt Mapped Read Duplication Rate: **11.19 (%) (excellent)**
- tumor_dna_aligned_metrics.txt Mapped Read Duplication Rate: **11.34 (%) (excellent)**
- Relatedness: **0.772 (very concerning for primary patient tumors; however low relatedness is common for highly aneuploid tumor cell lines)**
- normal.VerifyBamId.selfSM Contamination: **0.00025 (good)**
- tumor.VerifyBamId.selfSM Contamination: **0.00345 (good)**
- The proportion of RNA reads mapping to cDNA sequence is **0.84** (coding (0.49) + UTR (0.35)) **(good)**
- trimmed_read_1strandness_check.txt: Data is likely RF/fr-firststrand
- YAML file: immuno.strand: "first"
- Total Number of somatic variants called: 1,264

- Visual inspection of the plot below suggests no end bias in RNA-seq data



FDA Quality Thresholds

The following subset of QC values can be obtained from the FDA report tables generated by the pipeline and used to determine whether the case meets basic data quality criteria as described in documentation provided to the FDA.

Criteria	Threshold for pass/failure	File	Value	Pass
Contamination Estimate	<7.5%	normal DNA	0.00025	PASS
Contamination Estimate	<7.5%	tumor DNA	0.00345	PASS
Genotype Concordance	>95%	normal DNA	0.772	FAIL
Genotype Concordance	>95%	tumor DNA	0.772	FAIL
MEAN_INSERT_SIZE	125 - 300bp	tumor DNA	245.61	PASS
MEAN_INSERT_SIZE	125 - 300bp	normal DNA	249.73	PASS
MEAN_TARGET_COVERAGE	>100x	normal DNA	98.67	FAIL
MEAN_TARGET_COVERAGE	>250x	tumor DNA	123.87	FAIL
PCT_EXC_OFF_TARGET	<60%	tumor DNA	0.59	PASS
PCT_EXC_OFF_TARGET	<60%	normal DNA	0.60	PASS
PCT_PF_READS_ALIGNED	>95%	tumor DNA	0.998	PASS
PCT_PF_READS_ALIGNED	>95%	normal DNA	0.999	PASS
PCT_PF_READS_ALIGNED	>85%	tumor RNA	0.939	PASS
PCT_READS_ALIGNED_IN_PAIRS	>95%	normal DNA	0.999	PASS
PCT_READS_ALIGNED_IN_PAIRS	>95%	tumor DNA	0.999	PASS
PCT_TARGET_BASES_20X	>95%	tumor DNA	0.872	FAIL
PCT_TARGET_BASES_20X	>95%	normal DNA	0.869	FAIL
PCT_USABLE_BASES_ON_TARGET	>20%	normal DNA	25.05%	PASS
PCT_USABLE_BASES_ON_TARGET	>20%	tumor DNA	25.49%	PASS
PERCENT_DUPLICATION	<40%	tumor DNA	11.4%	PASS
PERCENT_DUPLICATION	<40%	normal DNA	11.2%	PASS
PF_MISMATCH_RATE_1	<0.75%	normal DNA	0.002298	PASS
PF_MISMATCH_RATE_1	<0.75%	tumor DNA	0.002399	PASS
PF_MISMATCH_RATE_2	<1.00%	normal DNA	0.002937	PASS
PF_MISMATCH_RATE_2	<1.00%	tumor DNA	0.003152	PASS
TOTAL_READS	>100M	normal DNA	77,667,085	FAIL
TOTAL_READS	>200M	tumor RNA	185,492,339	FAIL
TOTAL_READS	>250M	tumor DNA	96,056,889	FAIL

Tumor type / driver variant review

Two obvious driver variants were found, two known to be relevant to breast cancer specifically

- FGFR1 - ENSP00000400162.2:p.Ser125Leu (100% VAF)
- TP53 - ENSP00000269305.4:p.Arg175His (98.8% VAF)

HLA allele review

Class I

Tool	sample_type	HLA_A-1	HLA_A-2	HLA_B-1	HLA_B-2	HLA_C-1	HLA_C-2
optitype	normal	A*29:02	A*29:02	B*45:01	B*82:02	C*06:02	C*06:02
optitype	tumor	A*29:02	A*11:01	B*08:01	B*45:04	C*07:01	C*06:02
phlat	normal	A*01:22N	A*29:02	B*45:01	B*45:01	C*06:02	C*06:02
phlat	tumor	A*29:02	A*11:01	B*08:01	B*45:01	C*07:01	C*06:02
hlahd	normal	A*29:02	A*29:02	B*45:01	B*45:01	C*06:02	C*06:02
hlahd	tumor	A*29:02	A*11:01	B*08:01	B*45:01	C*07:514	C*06:127
clinical	normal	A*29:02	A*29:02	B*45:01	B*82:02	C*06:02	C*06:02

Class I alleles are usually much more concordant between algorithms, clinical typing, and normal/tumor samples than what is observed here. Since these are highly aneuploid cell lines with multiple mutations, it is likely that the tumor cell line has changed substantially from its corresponding normal sample. Here, since all the alleles had clinical HLA typing results, the pipeline defaults to using those. If clinical HLA typing results are not available, the Optitype algorithm's results are used for class I predictions.

- Confirmed that these class I alleles are what were used for runs of pVACseq/pVACview.
 - HLA-A*29:02,HLA-A*29:02,HLA-B*45:01,HLA-B*82:02,HLA-C*06:02,HLA-C*06:02

Class II

Tool	sample_type	DQA1-1	DQA1-2	DQB1-1	DQB1-2	DRB1-1	DRB1-2	DRB3-1	DPB1-1	DPB1-2	DPA1-1	DPA1-2
phlat	normal	DQA1*03:03	DQA1*03:03	DQB1*03:02	DQB1*03:02	DRB1*04:05	DRB1*04:05					
phlat	tumor	DQA1*03:03	DQA1*05:01	DQB1*02:01	DQB1*03:02	DRB1*03:01	DRB1*04:05					
hlahd	normal	DQA1*03:03	DQA1*03:03	DQB1*03:02	DQB1*03:02	DRB1*04:05	DRB1*04:05		DPB1*01:01	DPB1*01:01	DPA1*02:01	
hlahd	tumor	DQA1*03:03	DQA1*05:01	DQB1*02:01	DQB1*01:01	DRB1*03:01	DRB1*04:05	DRB3*01:01	DPB1*02:01	DPB1*01:01	DPA1*01:03	DPA1*02:01
clinical	normal	DQA1*03:01	DQA1*05:01	DQB1*02:01	DQB1*03:02	DRB1*03:01	DRB1*04:04	DRB3*01:01	DPB1*02:01	DPB1*04:01	DPA1*01:03	

Class II alleles are also usually much more concordant between algorithms, clinical typing, and normal/tumor samples than what is observed here. However, they do have a higher chance of being different within the in-silico prediction algorithms, and compared to clinical HLA typing. Since all the alleles had clinical HLA typing results, the pipeline defaults to using those.

- Confirmed that these class II alleles are what were used for runs of pVACseq/pVACview.
 - DQA1*03:03,DQA1*03:03,DQB1*03:02,DQB1*03:02,DRB1*04:05,DRB1*04:05,DPA1*02:01,DPB1*01:01,DRB4*01:03

Compare variant results to validated variants from a CLE pipeline.

CLE variants for comparison:

If exome sequence results were provided by a CLIA/CAP certified diagnostic lab (or similar) a comparison of somatic variants from the immuno pipeline used for neoantigen prediction should be compared to these orthogonal results.

Review multi-algorithm support for strong binding affinity.

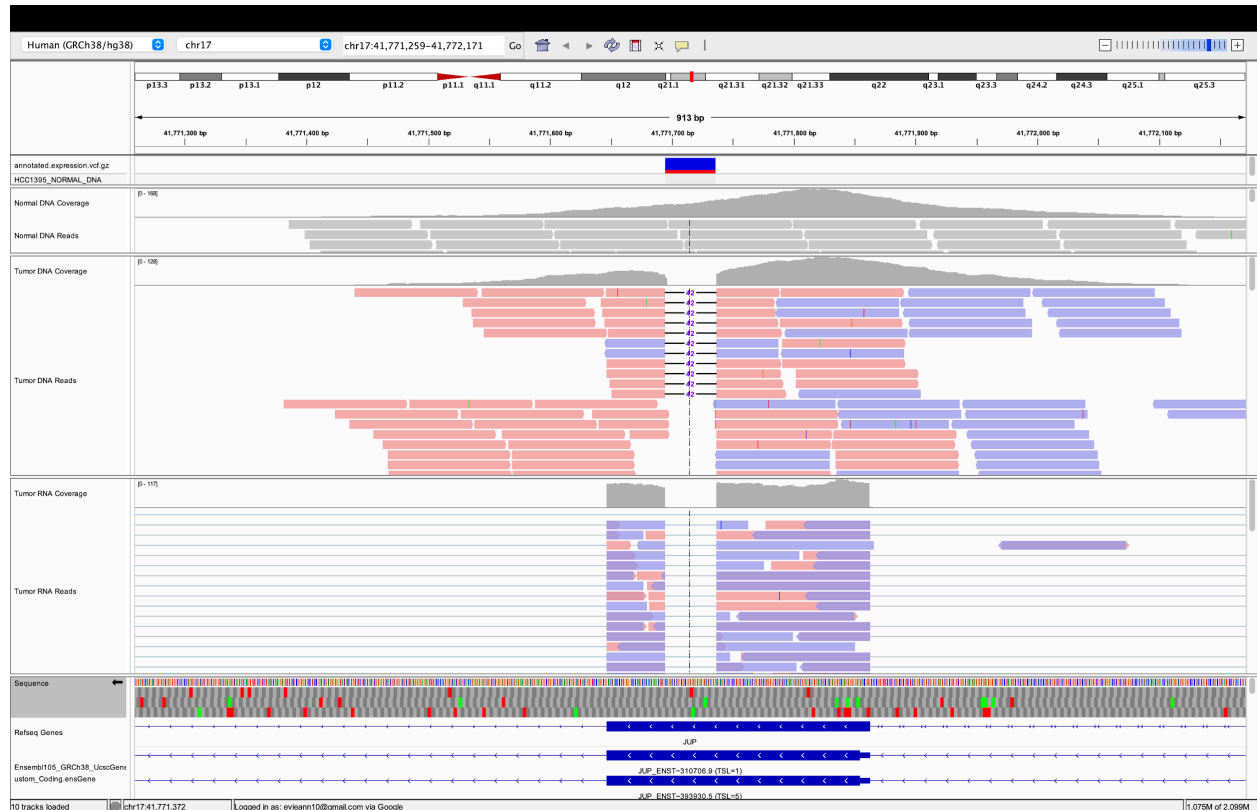
Notes on individual candidates are captured in the neoantigen candidates spreadsheet (**Supplementary Table 3**).

IGV Review

An IGV session was created to examine all 318 putative neoantigen candidates in the context of normal exome DNA alignments, tumor exome DNA alignments, tumor RNA-seq alignments, and transcript annotations used for peptide prediction (Ensembl v105). 240 candidates failed due to QC issues. Notes and illustrative IGV screenshots summarizing these candidates are provided below:

Mutation Not Called

JUP



Aggregate Report of Best Candidates by Variant

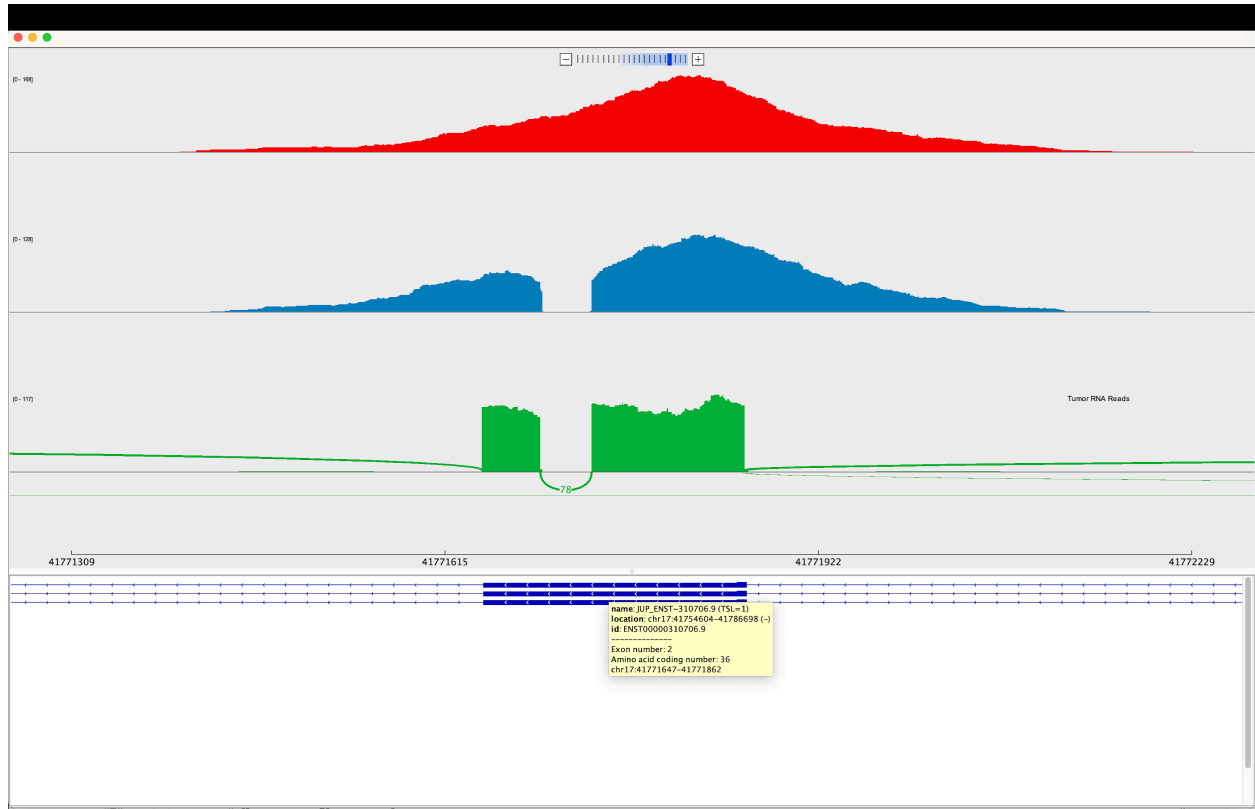
Column visibility ⌵ Search: JUP

	Gene	AA Change	Best Peptide	TSL	Allele	Pos	Prob Pos	Num Included Peptides	Num Passing Peptides	IC50 MT	IC50 WT	%ile MT	%ile WT	RNA Expr	RNA VAF	Allele Expr	RNA Depth	DNA VAF	Tier	Ref Match	Acpt	Rej	Rev
224	JUP	KGIMEDAEAGRQGR740-54T	SLKKTKTY	1	HLA-A*29:02	1	None	4	1	379,975	65,863	0.79	0.205	7.210	0.000	0.000	5	0.224	NoExpr	False			

Showing 1 to 1 of 1 entries (filtered from 319 total entries) Show 10 entries Previous 1 Next

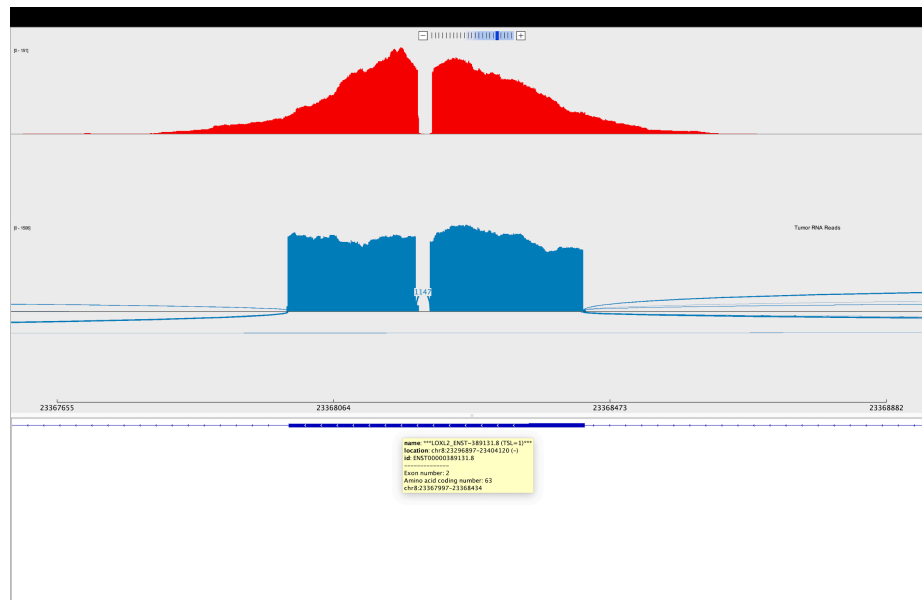
Currently investigating row:

224



Deletion of 42 bases looks real in DNA and RNA (masked in DNA due to softclipping). RNA deletion obviously there by read pattern even though the deletion is not called (manually filled in RNA VAF/expression values using a RNA depth of 100 which is average for the exon). No proximal variants. No unexpected RNA splicing. The majority of reads support the chosen transcript.

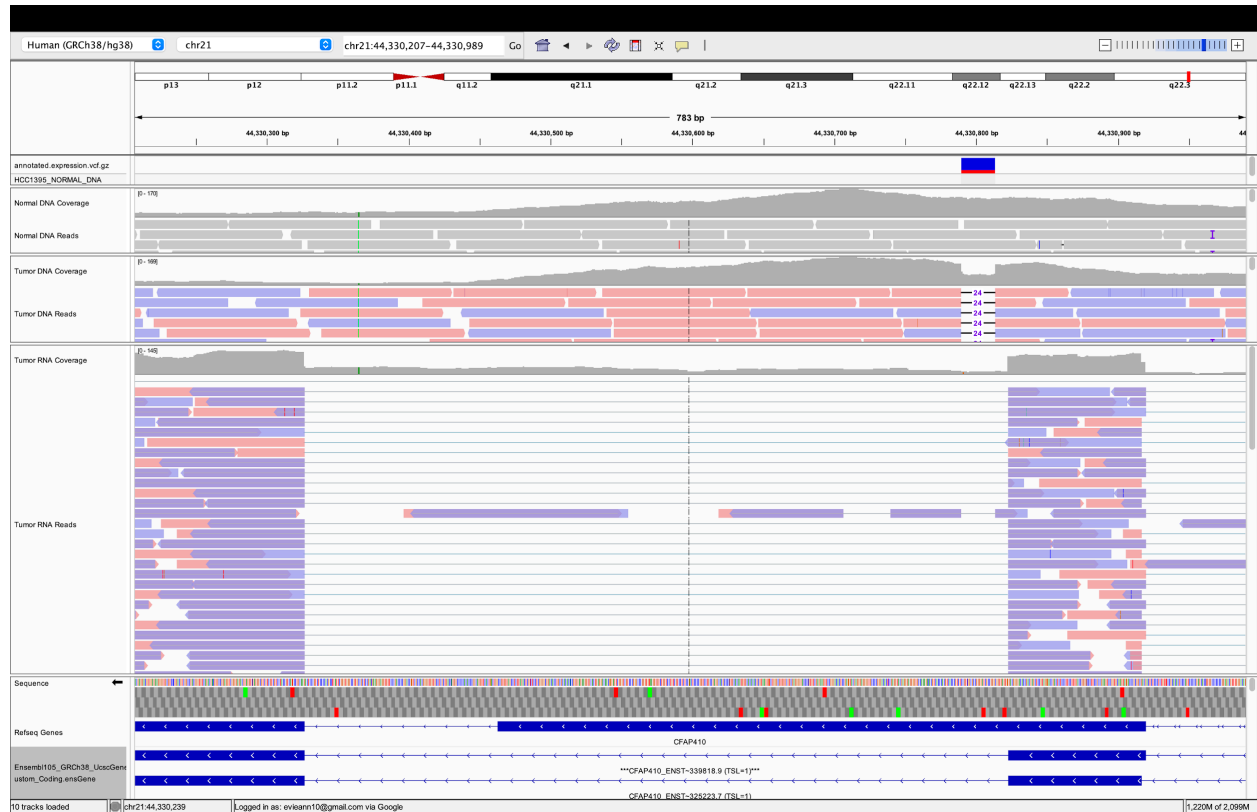
LOXL2



Deletion of 21 bases looks real in DNA but there is no deletion called in the RNA. There are reads that splice to one another in the same pattern as the deletion -- suggesting the deletion can be observed in the RNA. There is a SNV right after the deletion in the RNA which is likely a result of a misalignment of the deletion. No exon skipping. The majority of reads support the chosen transcript.

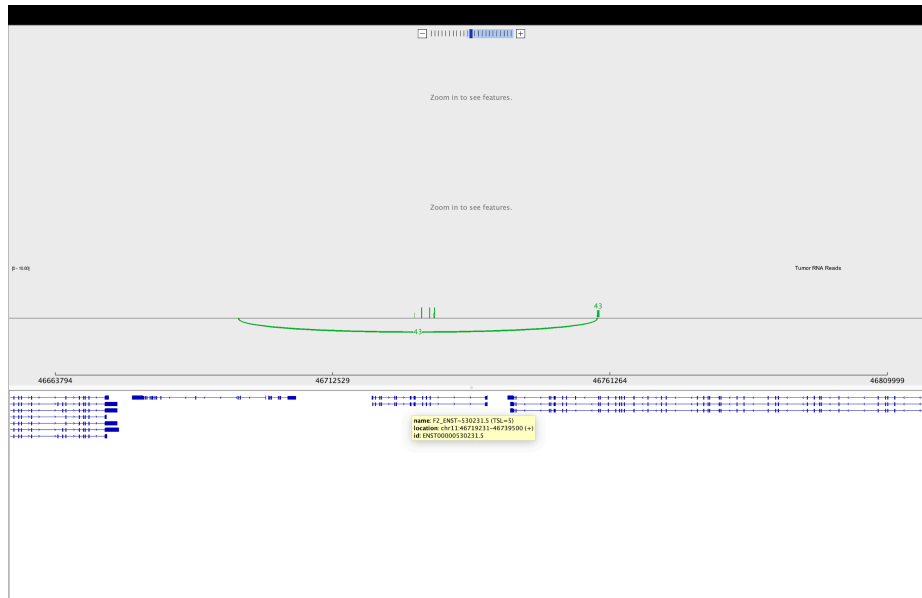
Incorrect Transcript

CFAP410



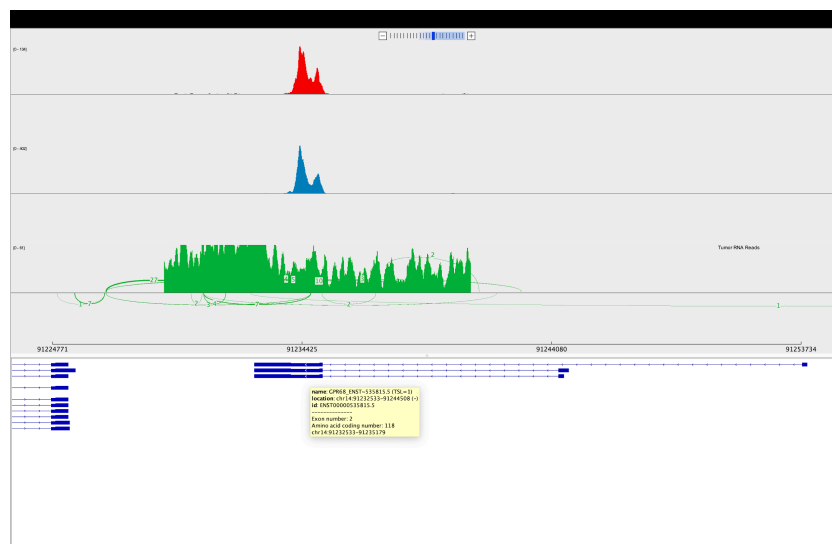
Deletion of 24 bases looks real in DNA but there are no called reads of evidence in the RNA. The RNA does not support the chosen transcript at all. No proximal variants that affect the 51mer. Rejecting for no RNA evidence.

F2



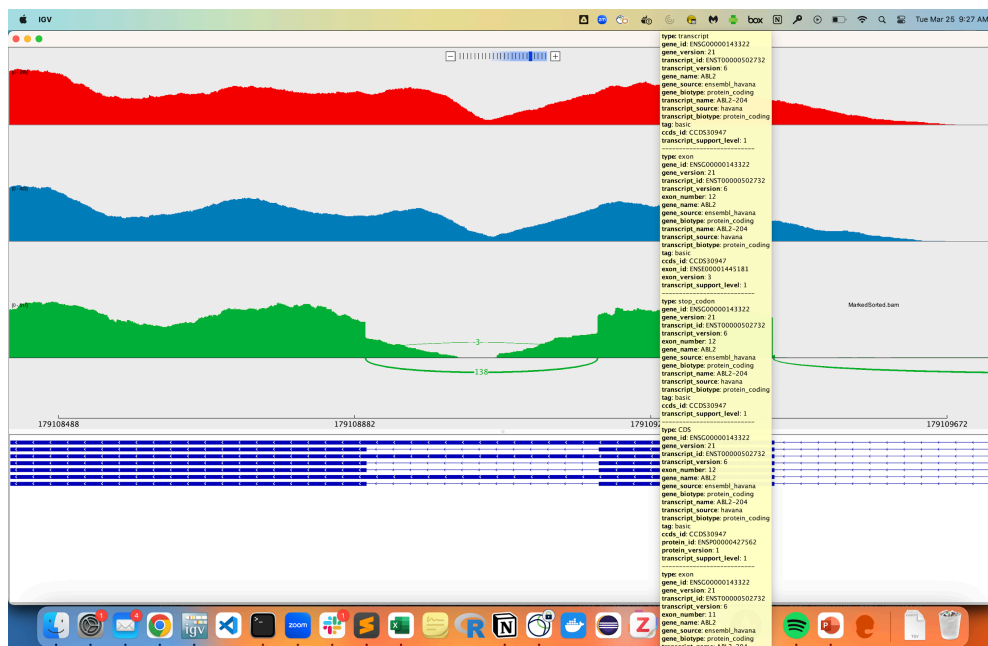
Somatic variant looks real in DNA but is in a region of low coverage in RNA (2 reads with variant are found in RNA). 1 proximal germline variant, results in no AA change. RNA reads do not support the chosen transcript, sashimi plot shows that the chosen transcript is not being utilized. Candidate is rejected due to lack of RNA support.

GPR68



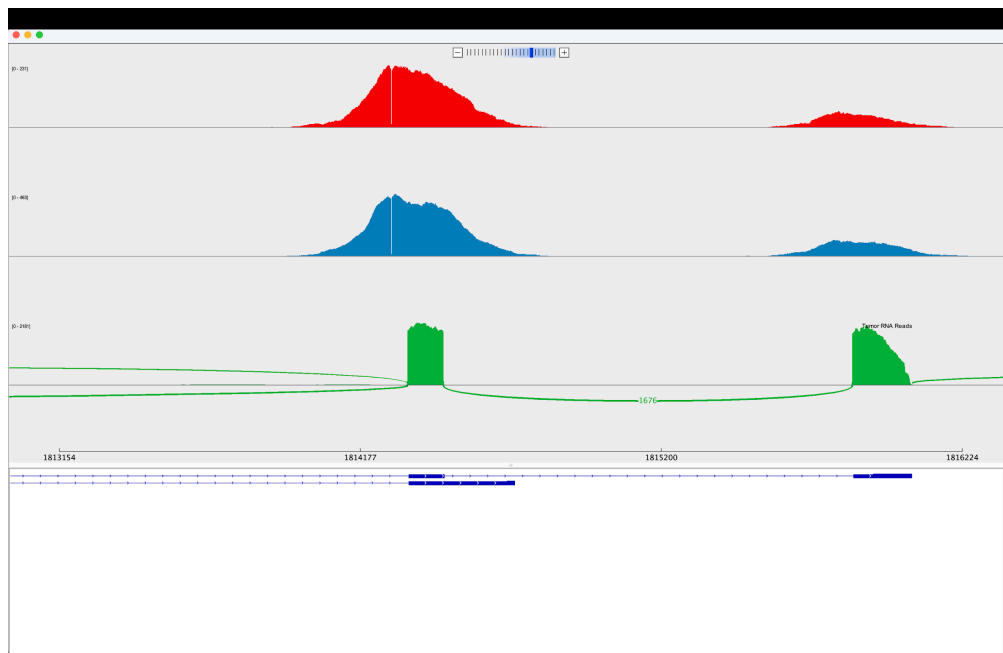
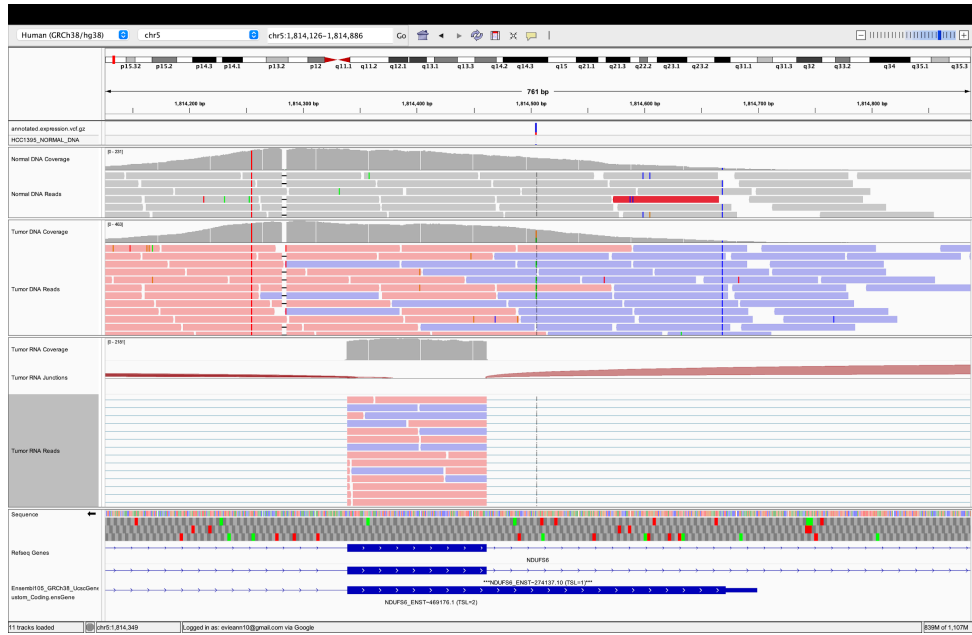
Deletion of 20 bases looks real in DNA but has 1 read supporting that variant in RNA despite being in a region with a lot of coverage. Abnormal splicing pattern/there are unannotated transcripts in the vicinity. Can't figure out if transcript annotation is correct.

ABL2



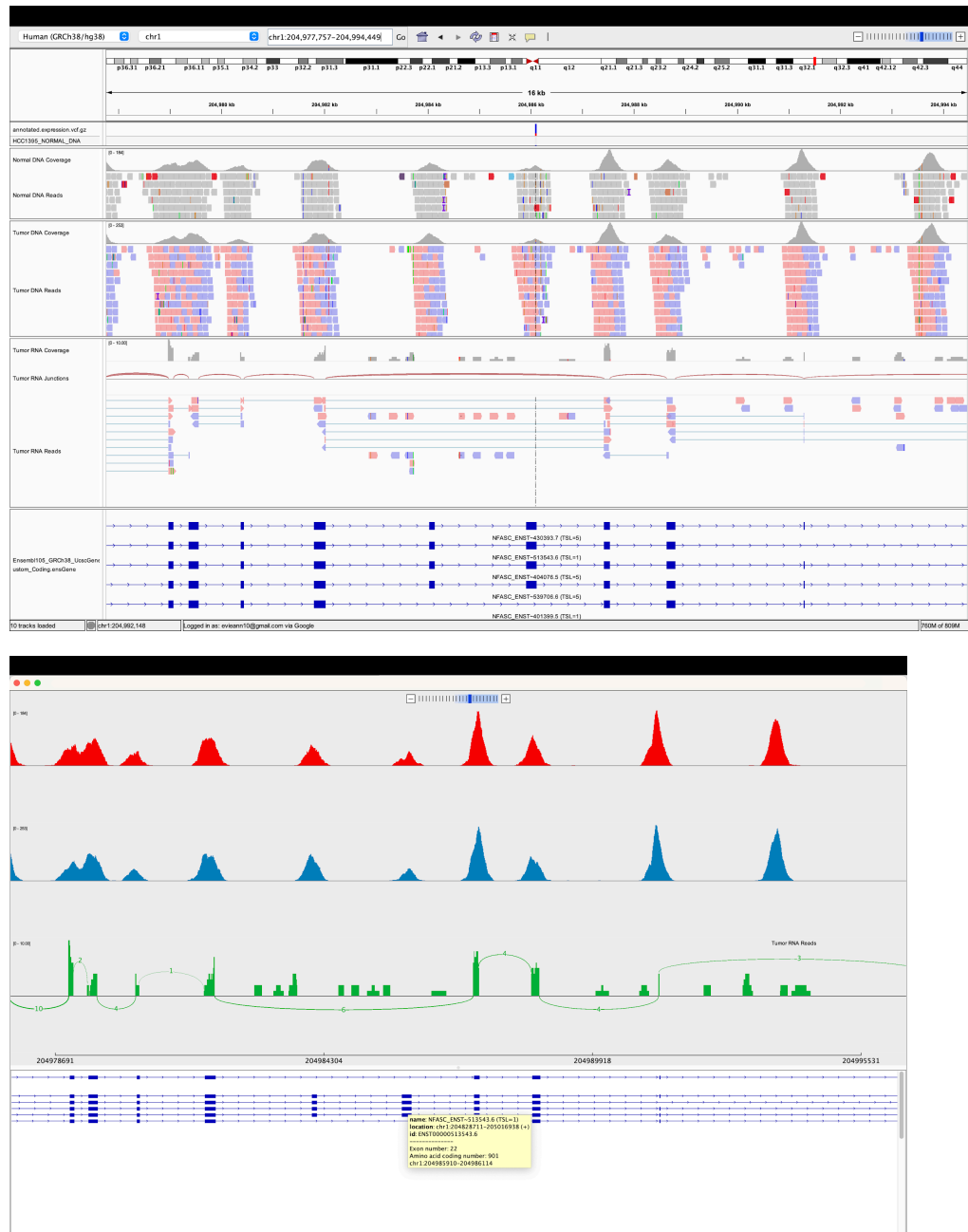
Somatic variant looks real in DNA and RNA. No proximal variants. Unfortunately, this transcript may be incorrect. The variant is found in what several other transcripts call an intron. The chosen transcript has an exon spanning through the intron but visually it doesn't seem that the read supports this transcript.

NDUFS6



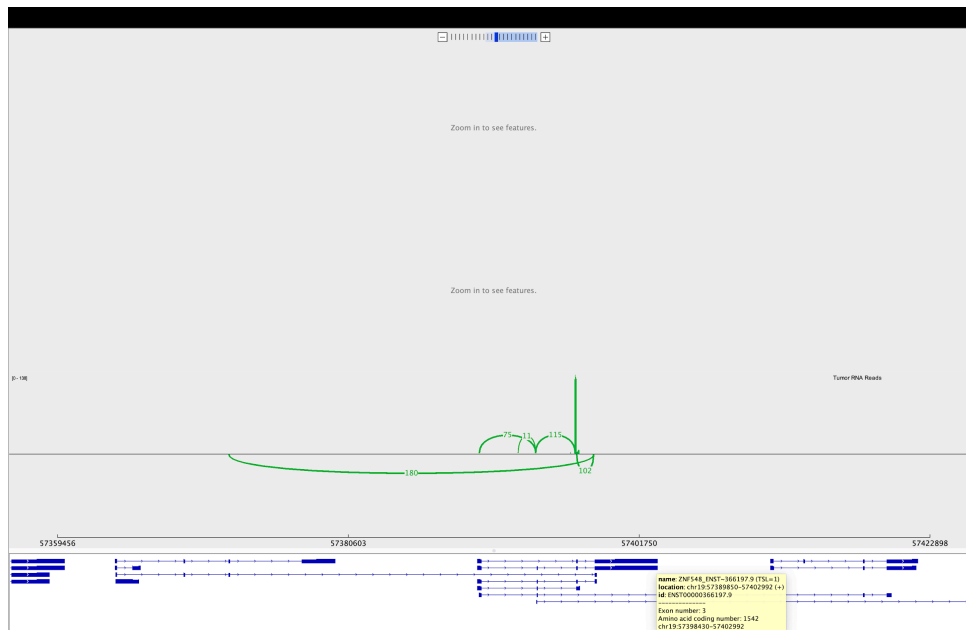
Somatic variant looks real in DNA but has very low support in RNA. No proximal germline variants. A majority of reads support another transcript, we will have to reject.

NFASC



Somatic variant looks real in DNA but there is no RNA coverage of this region -- we have to reject on the basis of no evidence of expression. It appears that there are other transcripts that show this region as an intron.

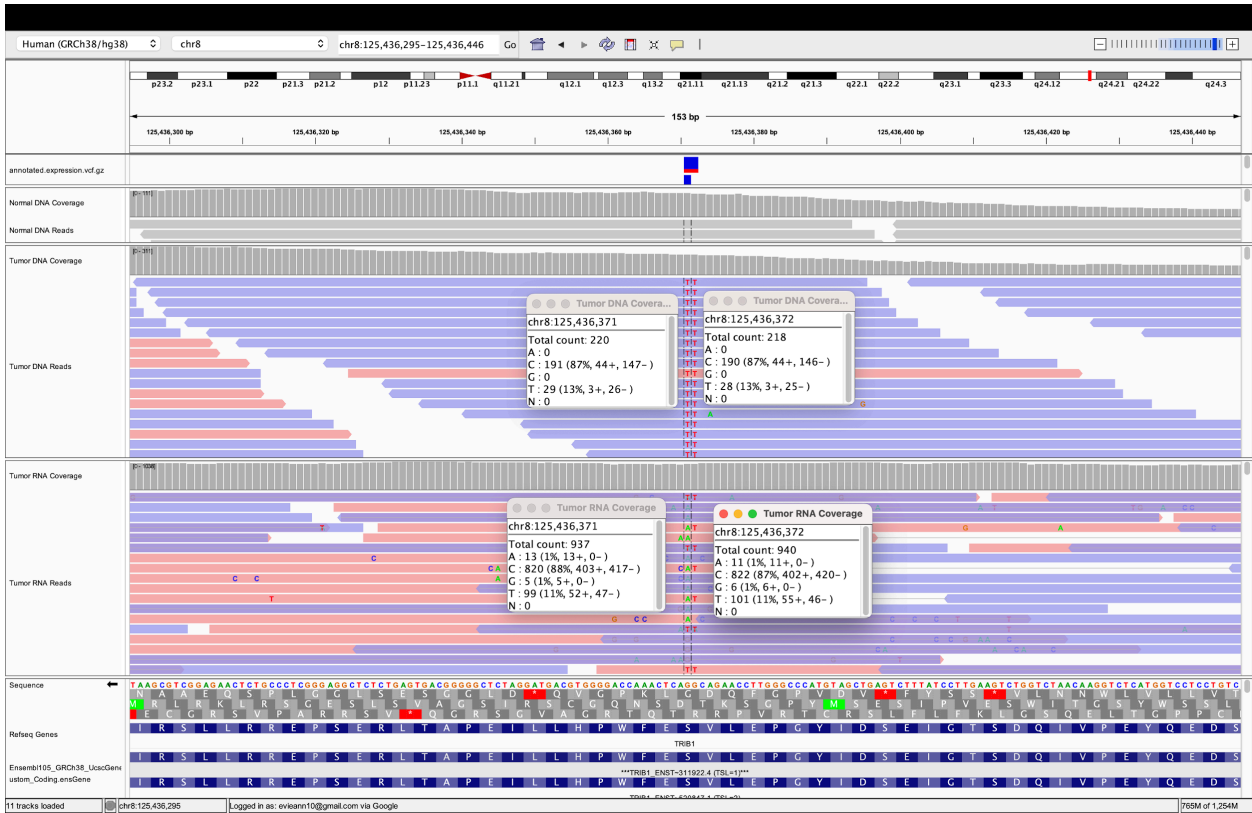
ZNF548



Somatic variant looks real in DNA and RNA. 51-mer, classII span upstream exon. 51-mer spans downstream exon. No proximal variants. There is an alternative isoform expressed which results in many reads skipping the exon entirely, still several reads support this exon and it has good coverage. I think we can accept this candidate.

Dinucleotide Miscall

TRIB1

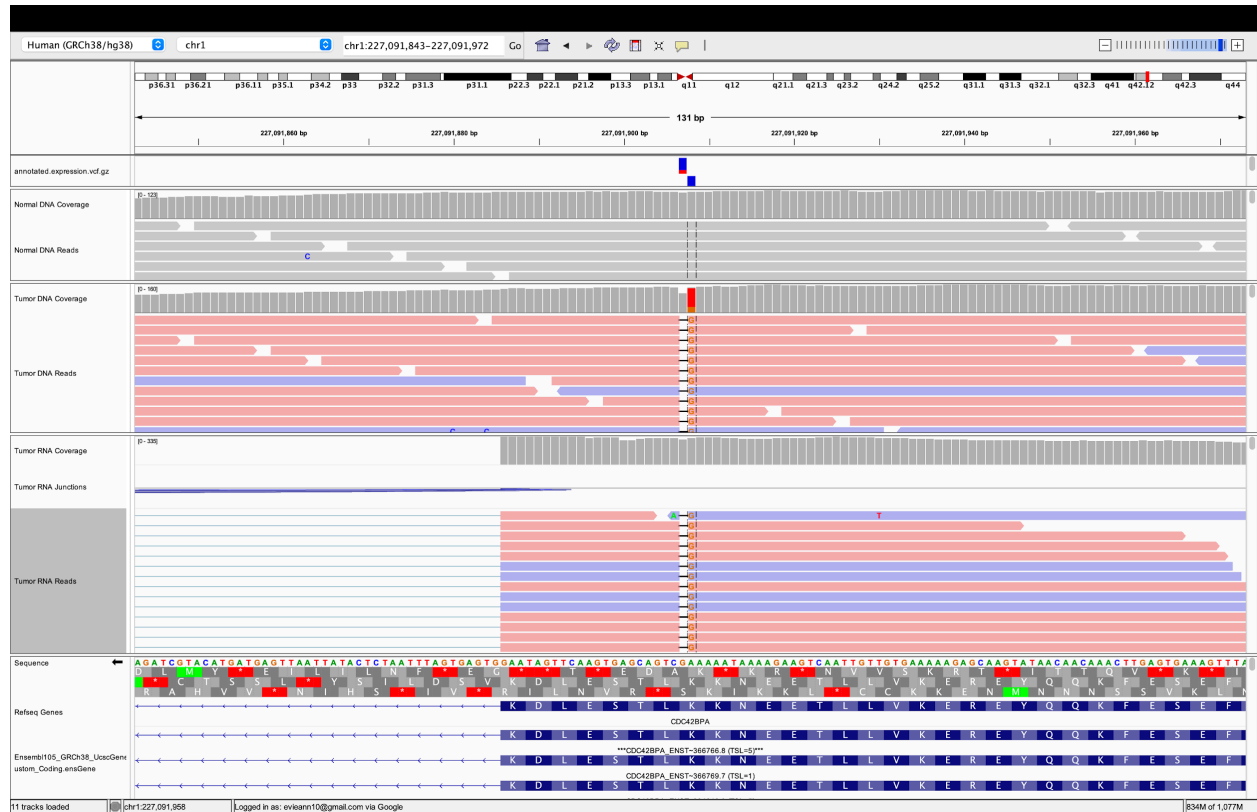


Aggregate Report of Best Candidates by Variant																			
Gene	AA Change	Best Peptide	TSL	Allele	Pos	Prob Pos	Num Included Peptides	Num Passing Peptides	IC50 MT	IC50 WT	%ile MT	%ile WT	RNA Expr	RNA VAF	Allele Expr	RNA Depth	DNA VAF	Tier	Ref Match
27 TRIB1	S340F	WFEFVLEPGY	1	HLA-A*29:02	4	None	8	3	107.97	111.208	0.5	0.432	78.490	0.102	8.008	795	0.133	Subclonal	False
33 TRIB1	S340F	WFEFVLEPGY	1	HLA-A*29:02	4	None	8	3	107.97	111.208	0.5	0.432	78.490	NA	NA		0.136	Subclonal	False

ClassI is great and ClassII is okay. DNA VAF is low and RNA VAF and allele expression are not shown because of dinucleotide miscounting (manually filled in using the lower counts). Not an anchor. The majority of algorithms agree it's a good binder. Elution scores are good. Dinucleotide variant entry will be accepted while the extra entries for this variant will be rejected.

Complex Variant

CDC42BPA



This somatic variant always happens in phase with a C deletion right before it (see candidate above this one). The deletion combined with the SNV results in the same AA change BUT the deletion causes a frameshift where the next AA is a stop codon, therefore I think we cannot accept this candidate because only half of the classI peptide would be captured in the 51mer. The SNV is the better binder so if we cannot target this we have to reject.

TRIM69



The 1-base insertion is in phase with an 18-base deletion immediately upstream (the next variant). Somatic variants look real in DNA and RNA. No unexpected RNA splicing. The majority of reads support the chosen transcript. Can accept but will need to confirm if the peptide generated accurately reflects both mutations.

Reference Match

FGFR1

Aggregate Report of Best Candidates by Variant

Column visibility

Search: FGFR1

	Gene	AA Change	Best Peptide	TSL	Allele	Pos	Prob Pos	Num Included Peptides	Num Passing Peptides	IC50 MT	IC50 WT	%ile MT	%ile WT	RNA Expr	RNA VAF	Allele Expr	RNA Depth	DNA VAF	Tier	Ref Match	Acpt	Rej	Rev
41	FGFR1	S36L	LPEQDALPSL	1	HLA-B*82:02	10	None	1	0	766.27	11959.820	0.3	9.500	143.961	0.999	143.817	1501	1.000	Poor	True			

Showing 1 to 1 of 1 entries (filtered from 319 total entries) Show 10 entries

Currently investigating row: 41

Previous 1 Next

Variant Information

Transcript Sets of Selected Variant

Reference Matches

Additional Data

Best Peptide Data

Best Peptide: LPEQDALPSL

AA Change: S36L

Pos: 10

Gene: FGFR1

Query Data

Query Sequence: EDDALPSLEDDDDDD

Hits: 4

Hits

Show 10 entries

Search:

Matched Peptide	Genes	Transcripts	Hit IDs
1 SLEDDDD	DDX46	ENST00000354283.8, ENST00000452510.7, ENST00000507392.5, ENST00000628477.2	ENSP00000487025.1, ENSP00000427290.1, ENSP00000416534.2, ENSP000003462

Showing 1 to 1 of 1 entries

Previous 1 Next

Variant & Gene Info

DNA VAF: 1

RNA VAF: 0.999

Gene Expression: 143.961

Genomic Information (chromosome - start - stop - ref - alt): chr8-38428419-38428428-G-A

Additional variant information: [OpenCRAVAT variant report](#)

Peptide Evaluation Overview

Evaluation	Count
Pending	319

The last 2 amino acids of the best peptide along with the following sequence perfectly matches a wild-type peptide sequence in a different gene. Therefore, it is unlikely this peptide will be immunogenic, and it should be rejected.

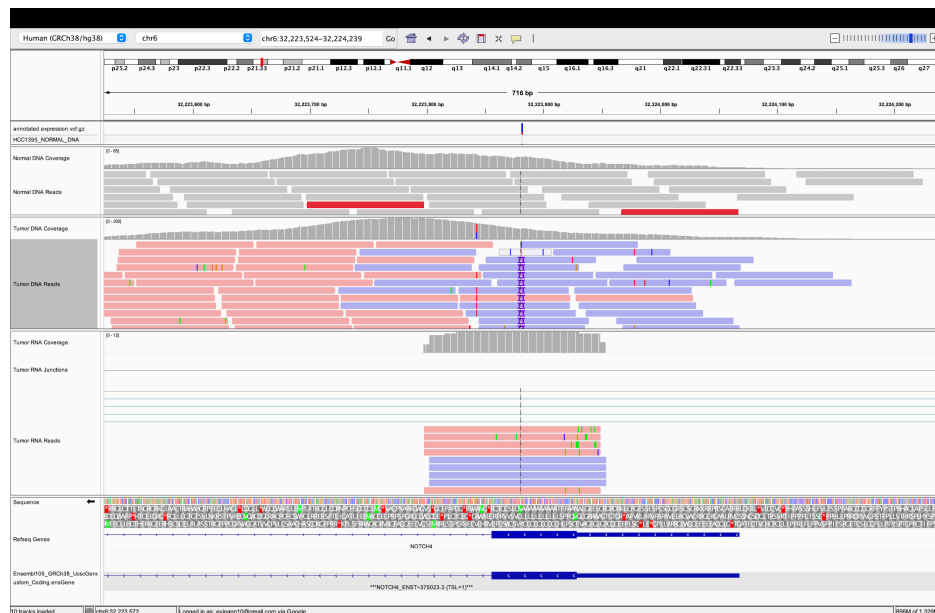
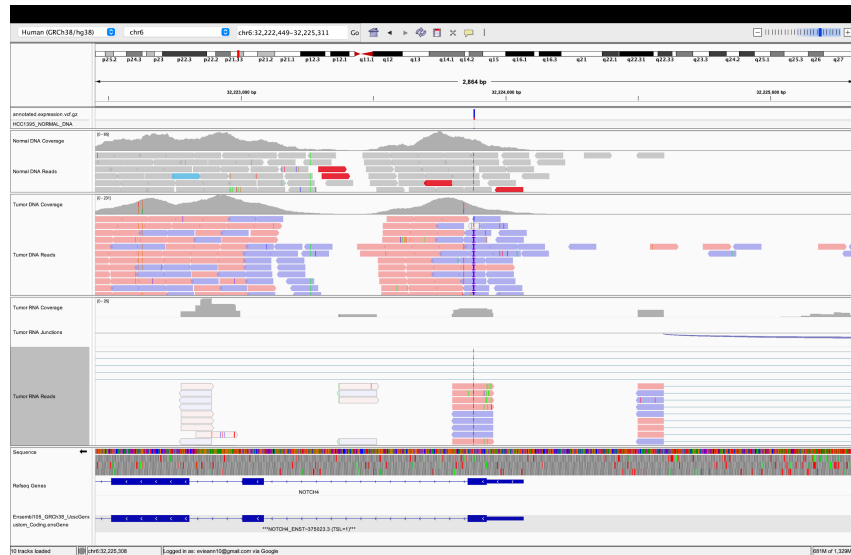
SERPINA1



Somatic variant looks real in DNA and RNA. 1 proximal germline variant downstream of variant of interest - results in a R->H which was properly accounted for in the 51mer. 51mer spans 1 exon. No unexpected RNA splicing. The majority of reads support the chosen transcript.

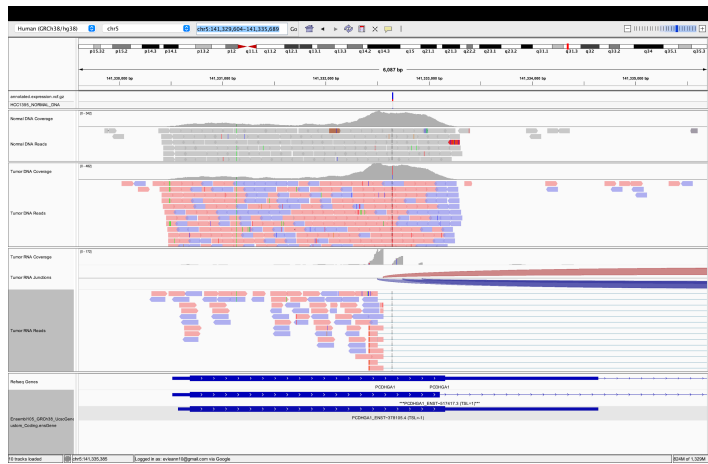
Insufficient RNA expression

NOTCH4



The somatic variant does not look real; The noted insertion could be a misalignment due to the nearby repetitive soft-masked region. Additionally, in the RNAseq data, most of the reads aligning to this gene are either multi-mapped reads, or align to the introns, further raising doubts about this region. Rejecting due to inconclusive evidence of variant in DNA and lack of expression in RNA.

PCDHGA1



The somatic variant looks real in DNA but is in a region of poor coverage in RNA. This gene is part of a family of single-exon genes with high sequence similarity that are prone to incorrect mapping, resulting in a noisy sashimi plot. Rejecting this candidate due to insufficient evidence in RNA.